

The COVID-19 Pandemic Shock and Life Satisfaction of Chinese Residents: The Roles of Disease Burden and Lockdown Stringency

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Abstract

This study examines how disease severity and containment policy stringency were associated with changes in life satisfaction among Chinese residents during the COVID-19 pandemic. I combine seven waves of the China Family Panel Studies from 2010 to 2022 with province level indicators from the Oxford COVID-19 Government Response Tracker. The analytical sample includes 17,733 person wave observations from 3,825 adults across 25 provinces. I use a continuous treatment difference in differences framework with individual fixed effects and region by wave fixed effects to estimate within person changes linked to differences in provincial disease burden and containment exposure.

The results show a clearer negative association for cumulative disease burden than for containment duration. In the preferred joint specification, logged cumulative disease burden is associated with lower life satisfaction, with $\beta = -0.0331$, $p = 0.009$, and a wild cluster bootstrap p value of 0.019. Strict containment duration also has a negative coefficient, but the estimate remains statistically imprecise, with $\beta = -0.0172$ and $p = 0.213$. Standardised estimates do not show a significant difference in magnitude between the two exposure channels. Wave specific estimates suggest a stronger disease relationship in 2020 and a more negative containment pattern in 2022, although formal differences between waves remain imprecise. Heterogeneity analysis suggests that negative containment associations are more evident among younger respondents, people with non-agricultural hukou, and respondents with at least lower secondary education.

The findings suggest that pandemic related changes in subjective wellbeing varied across exposure types, periods, and population groups. Separating disease burden from policy stringency within the same longitudinal framework provides a more differentiated account of life satisfaction during the pandemic in China. The evidence is more consistent for cumulative disease burden, while province level treatment assignment, limited independent clusters, exposure measurement error, and imperfect temporal ordering constrain strong causal interpretation.

Keywords: COVID-19; life satisfaction; containment policy stringency; continuous difference in differences; China

Acknowledgements

I once imagined that, when the time came to write these acknowledgements, the words would pour out effortlessly. Yet when I finally reached this page, I found myself sitting in silence for a long while.

I suppose you, too, must have travelled a long way to arrive here.

To encounter a truly inspiring mentor is one of life's greatest fortunes. No number of words could ever fully capture the depth of my gratitude.

Looking back at the lines of feedback you wrote and the voice messages you sent, your advice was always concise, yet remarkably perceptive, often reaching directly to the heart of the problem.

I may not have been the most academically outstanding student you have ever supervised, but I have always treated every piece of your advice and every lesson you offered with the utmost seriousness. In countless quiet nights, I worked a little harder, reflected a little more deeply, and tried, step by step, to become better than I had been before.

My deepest gratitude also goes to the friends who, even on the darkest nights, still rang the doorbell with warmth and light.

Together, we have witnessed one another's happiest moments and some of the most important turning points in our lives. Across distance, time, crowds, departures, and reunions, these moments have accumulated into memories of friendship that belong to us alone.

I often wish that some of us had met sooner. Yet perhaps certain encounters were always meant to happen when they did.

Finally, I would like to thank myself—the person who is still moving forward, still trying, still capable of being delighted by small things and moved by kindness.

I am grateful for the person who has finally learned how to cherish what she has and reflect on who she is; for the person who is no longer afraid of being ordinary.

We are always saying goodbye, and we are always on our way somewhere. Sometimes we stop and wonder whether we have taken a wrong turn, whether the road beneath our feet still leads to where we once intended to go.

But perhaps life does not always have to follow the route marked out by a map.

I hope that, when one day we finally reach one of those high places in life and look back at everything behind us, we will remember to raise a glass to the younger versions of ourselves—to thank them properly for having come this far.

In the year ahead, may we all continue moving, little by little, towards better places and better versions of ourselves.

To borrow a few words from *The Curious Case of Benjamin Button*:

“I hope you live a life you’re proud of.

I hope you see things that startle you.

I hope you feel things you never felt before.

And if you find that you’re not,

I hope you have the courage to start all over again.”

For myself at twenty-three,

and for every version of myself still waiting in the years ahead.

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1 Introduction

1.1 Research Background

COVID-19 disrupted daily life across the world on a scale few recent events match. Diener (1984) framed subjective well-being (SWB) as three components, cognitive life satisfaction together with positive and negative affect. Researchers since then have relied on life satisfaction as the main gauge of how people judge their lives as a whole (Diener, Emmons, Larsen and Griffin, 1985; Diener, Suh, Lucas and Smith, 1999). I find this measure useful for reading the human cost of the pandemic, since case counts and death tolls say little about how people actually felt.

Evidence gathered across many countries points the same way. The pandemic pushed life satisfaction down, though the size of the drop differed by country and shifted over time. Aknin et al. (2022), drawing on longitudinal data from 15 countries, reported that tighter government responses tracked worse mental health. Brodeur et al. (2021) reached a similar reading through Google Trends, where lockdowns coincided with sharp rises in searches for boredom, loneliness and worry. Easterlin and O'Connor (2023) traced European life satisfaction moving almost in step with each pandemic wave, recovering to pre-pandemic levels only in 2022.

Comparative work sharpens the picture. Greyling, Rossouw and Adhikari (2021) linked lockdown stringency to lower happiness, while Caro et al. (2022) associated stricter lockdowns with greater loneliness across European settings. Grimes (2022) found that more stringent lockdowns were associated with lower wellbeing. Cross, Ng and Scuffham (2020) added an economic angle, showing that stricter responses carried measurable costs without delivering proportionally better epidemic control.

I read these findings as leaving one question unsettled. How much of the wellbeing loss traces back to the disease, and how much to the containment policies, above all lockdowns, adopted against it? The answer carries normative weight. Losses driven mainly by disease severity would make restrictive policy a defensible trade-off, whereas losses driven mainly by lockdowns would invite a rethink of the policy calculus.

China offers a setting well suited to separating these two channels. It faced the outbreak first and imposed highly stringent containment measures early in the pandemic (Cross, Ng and Scuffham, 2020), while disease severity and lockdown intensity still varied widely

across its 31 provinces. That internal variation makes it possible to pull apart the contribution of illness from the contribution of policy to shifts in life satisfaction, which cross-national comparisons rarely allow.

Chinese studies have already opened this ground. Zheng, Miao and Gan (2020) matched provincial epidemic data to individual life satisfaction and general health across all 31 affected provinces. Ma, Wang and Wu (2024) applied a difference-in-differences design to happiness expressed on Weibo, separating the effect of outbreak news from the Wuhan lockdown itself, and found a protective lockdown effect that faded as time passed. Zhang, Battisti and Proto (2024) worked with nationally representative CFPS panel data and showed sharper mental health decline among people under stricter lockdowns. Wei et al. (2022) recorded shifts in trust, perceived fairness and wellbeing before and after the outbreak began. Huang et al. (2022) showed that regional pandemic severity was associated with differences in perceived life satisfaction, while Huang et al. (2025) examined how policy stringency related to patient satisfaction with community pharmacies. Charpe (2022) found province-level stringency depressing local economic activity independently of case counts, which suggests that stringency works through channels separate from disease exposure.

What remains thin is a unified framework that decomposes and compares the two forces across Chinese provinces. Few studies weigh lockdown stringency against disease severity within one design. The gap matters theoretically, since it bears on debates about the tripartite structure and drivers of subjective wellbeing (Diener, Suh, Lucas and Smith, 1999). It matters practically as well. Governments still calibrate pandemic responses and prepare for the next public health emergency, so knowing whether wellbeing costs come more from the pathogen or from the response should shape interventions that protect health and happiness together.

I therefore examine how variation in pandemic severity and lockdown stringency across Chinese provinces shaped residents' life satisfaction. The study aims to speak to the wellbeing literature and to inform how future pandemic policy might be designed.

1.2 Research Significance

Theoretically, this study contributes to research on subjective well-being by examining how the COVID-19 pandemic was associated with changes in life satisfaction at the individual level. Rather than treating the pandemic as a single and uniform shock, it distinguishes

between two forms of provincial exposure: disease burden and the stringency of containment policies. By linking these province-level conditions to longitudinal CFPS data, the study captures how individuals' life satisfaction changed under different levels of pandemic exposure. This approach adds a temporal and subnational perspective to existing research and further allows the analysis to examine whether these relationships varied across population groups.

Practically, distinguishing the respective roles of disease severity and policy restrictions can provide useful evidence for the design of future public-health responses. It highlights the importance of considering not only infection control but also the broader well-being consequences associated with prolonged restrictions. Identifying heterogeneous responses may also help policymakers recognise population groups that require greater social or psychological support during large-scale public-health emergencies.

1.3 Research Objectives

This study aims to quantify changes in life satisfaction during the pandemic and examine whether these changes were systematically associated with provincial differences in disease burden and policy stringency. Using longitudinal CFPS data with continuous-DiD models, it further investigates temporal and population heterogeneity in these relationships. More broadly, the study seeks to provide a differentiated account of how the health and policy dimensions of the pandemic were related to subjective well-being in China.

1.4 Structure of the Dissertation

The remainder of the dissertation is organised as follows. The literature review examines existing research on the pandemic, containment policies, and subjective well-being and develops the analytical framework. The methodology section describes the data, sample construction, variable measurement, and empirical strategy. The results section presents the main findings, robustness checks, and heterogeneity analyses. Finally, the conclusion summarises the findings, discusses their implications and limitations, and identifies directions for future research.

2 Literature Review

2.1 Scope and Conceptual Framework

COVID-19 simultaneously disrupted health, employment, social interaction and everyday routines, making it an unusually broad shock to subjective wellbeing. Life satisfaction is particularly relevant because it captures an evaluative judgement of life as a whole rather than a transient emotional state. Two related but conceptually distinct exposures are central to understanding this change. The first is disease burden, including infection risk, illness, mortality and pressure on health services. The second is the policy response, including restrictions on mobility, work and social interaction. Each may affect wellbeing through psychological, behavioural and economic pathways, and the magnitude of these effects may differ across population groups. A coherent review must therefore separate the disease and policy channels analytically while recognising that they are jointly determined in practice.

The evidence base evolved with the pandemic itself. Early studies relied largely on rapidly fielded surveys and concentrated on acute psychological responses. As repeated waves became available, longitudinal research began to examine within-person adaptation and divergence across groups. More recent studies have followed respondents into the recovery period and increasingly used panel or quasi-experimental designs to strengthen causal interpretation. This chronological progression matters analytically because changes in substantive conclusions often reflect improvements in data and research design rather than simple disagreement across studies. The review therefore first traces this methodological evolution, then synthesises the evidence around disease severity, policy stringency, mechanisms and heterogeneity. It concludes by identifying the measurement and identification problems that motivate the present province-level panel analysis.

2.2 Evolution of the Evidence

2.2.1 Early Cross-Sectional Evidence

The earliest studies were conducted when pre-pandemic baselines that could be linked to new COVID-19 surveys were rarely available. Their main contribution was therefore to document contemporaneous associations and develop plausible mechanisms rather than to identify changes caused by the outbreak. Clemente-Suárez et al. (2020) synthesised

early evidence on the pandemic's social and psychophysiological consequences and highlighted fear, isolation and economic anxiety as likely sources of deteriorating wellbeing. Schwinger et al. (2020) provided individual-level evidence from Germany, showing that the first lockdown was associated with lower satisfaction of autonomy, competence and relatedness and that these unmet psychological needs accounted for much of the deterioration in wellbeing, anxiety and depression. Zheng, Miao and Gan (2020) further showed that perceived control buffered the relationship between pandemic exposure, general health and life satisfaction, with psychological distance acting as a mediator.

These studies established that the pandemic should not be treated as an undifferentiated stressor. Objective exposure appeared to affect wellbeing partly through perceived threat, loss of control and disruption to basic psychological needs. Their main limitation was temporal identification: without observing the same respondents before the outbreak, pandemic responses could not be cleanly separated from stable differences in personality, resources or prior mental health. Even so, the mechanisms proposed in this early literature shaped subsequent longitudinal work. Perceived control implies that policy can generate both costs and reassurance, while psychological distance implies that the same disease burden can be experienced differently across places and groups. These insights provide the conceptual basis for distinguishing objective provincial exposure from individual responses in the present study.

2.2.2 Longitudinal and Panel Studies

The next phase broadened geographical coverage and introduced repeated observations. Alzueta et al. (2021) documented psychological correlates across 59 countries and showed substantial variation in pandemic experiences across national contexts. Bhoyroo et al. (2021) and Raj and Bajaj (2023) examined lifestyle, mental health and coping during lockdown in Australia and India; Raj and Bajaj (2023) identified living alone as a particular risk among young working adults. Ekici and Watson (2022) linked psychological wellbeing to consumer life satisfaction and policy, extending the outcome beyond clinical distress. Although not every study in this phase included a pre-pandemic baseline, the broader samples established an important point: the association between restrictions and wellbeing was conditioned by social position and institutional context rather than being uniform across populations.

Studies with repeated observations then shifted the question from whether wellbeing deteriorated to how it changed over time. Schlomann et al. (2022) followed older adults in Germany and found greater psychosocial resilience than early accounts had anticipated. Liu (2022) used three waves within one year to examine the shape of subjective-wellbeing change rather than relying on a single before-and-after contrast. Levacher et al. (2023) examined the beginning of the global outbreak, while Taylor et al. (2022) used qualitative evidence to show how individuals in the United Kingdom understood and managed lockdown. Gabet et al. (2023) demonstrated the value of purpose-built cohort infrastructure through the pan-Canadian COHESION study. Taken together, these studies displaced the simple narrative of universal decline with a more dynamic account: acute disruption was often followed by partial adaptation, but the timing and completeness of recovery varied markedly across individuals.

2.2.3 Longer Horizons and Stronger Identification

Later studies extended the observation window into the recovery period. Easterlin and O'Connor (2023) examined three years across Europe and found that life satisfaction had largely recovered in most countries by 2022-2023. Delgado-Rodríguez, Pinto Hernández and Tailbot (2024) reported a similarly modest and mainly temporary imprint in the Netherlands. Zrnić Novaković et al. (2025) followed mental health over 42 months in five European countries, while Nakagawa et al. (2025) distinguished rank-order stability from mean-level change in Japan. Phulkard et al. (2023) extended the geographical evidence to Thailand. Collectively, these studies support a broad pattern of long-run resilience. At the same time, they make clear that recovery in a national mean cannot be read as evidence of equal recovery across population groups.

The newest work combines longer follow-up with explicit heterogeneity analysis or stronger panel designs. Kung and Steptoe (2024) used an outcome-wide analysis among socially isolated older adults in the United Kingdom, reducing the selective-outcome problem that characterised some earlier research. Molla et al. (2026) modelled life-satisfaction trajectories before, during and after the outbreak, while Wang et al. (2026) examined educational convergence and the equalisation of happiness in China. Tuna et al. (2023) used piecewise linear panel models to represent non-linear, region-specific phases. Ma, Wang and Wu (2024) treated lockdown and infection as separable time-varying exposures in China, and Dui (2022) and Wan, Wang and Jiang (2023) used CFPS and provincial evidence to analyse wage and regional-income inequality. The methodological

trajectory is therefore clear: the literature has moved from documenting an aggregate shock, to estimating within-person change, and then to modelling heterogeneous trajectories and more credible counterfactual comparisons.

2.3 Two Exposure Channels and Their Mechanisms

2.3.1 Disease Severity

One strand of the literature treats cases, deaths or infection as indicators of objective disease severity. Huang et al. (2022) showed that country- or region-level pandemic severity moderated the relationship between personal risk experience, life satisfaction and psychological distress. Thompson et al. (2022), pooling 11 United Kingdom longitudinal studies, found that infection was followed by modest deterioration in psychological distress and life satisfaction. Their estimates were smaller than those reported in many cross-sectional comparisons, illustrating how within-person evidence can materially revise substantive conclusions. Sinha and Maiti (2025) also linked psychosocial status to COVID-19 outcomes, while Cui and Han (2022) and Li et al. (2024) documented mental-health burdens among Chinese residents associated with local pandemic conditions.

This evidence supports disease burden as a plausible determinant of wellbeing but leaves unresolved measurement problems. Confirmed cases reflect not only true transmission but also testing capacity, reporting rules and public-health surveillance. Infection may affect wellbeing directly through illness and indirectly through fear, service disruption or local economic decline. A province-level case rate is therefore an exposure proxy rather than an individual treatment. The literature also provides no basis for assuming that cumulative burden and recent incidence should relate to life satisfaction in the same way. Cumulative cases may better capture prolonged experience and accumulated memory, whereas new cases may better represent immediate risk. Comparing stock and flow measures is consequently a substantive robustness exercise rather than a choice between an obviously correct and an incorrect indicator.

2.3.2 Containment-Policy Stringency

A second strand focuses on restrictions rather than epidemiological severity. Grimes (2022) argued that pandemic and lockdown impacts should be measured separately because a policy response is not equivalent to the disease conditions that triggered it. Foliano, Tonei and Sevilla (2024) identified reduced leisure activity as a concrete pathway

through which social restrictions lowered wellbeing. van den Boom et al. (2023) followed mental health, loneliness and life satisfaction across two years of changing Dutch policy, while Henseke, Green and Schoon (2022) examined subjective wellbeing during the second pandemic phase, when emergency lockdown had evolved into prolonged restriction. Together, these studies suggest that the timing, duration and intensity of policy may matter as much as a binary distinction between lockdown and no lockdown.

The policy channel is theoretically ambiguous rather than uniformly harmful. Restrictions can reduce wellbeing through losses of work, mobility, leisure and social contact, yet they may also reduce fear by signalling protection and lowering infection risk. Krekel et al. (2023) further reversed the conventional direction of explanation by showing that happiness predicted compliance with preventive health behaviours. This creates an endogenous feedback loop: prior wellbeing may influence compliance, compliance may alter measured disease severity, and disease severity may in turn provoke stricter policy. A regression that includes disease and policy together can estimate the conditional association of each exposure given the other, but it does not automatically identify two independent causal mechanisms. The literature therefore supports a dual-exposure design while simultaneously requiring restraint in causal interpretation.

2.3.3 Psychological, Behavioural and Economic Mechanisms

Mechanism studies explain why identical objective exposure may produce different satisfaction responses. Negri et al. (2023) modelled meaning, beliefs and perceived vulnerability or mortality as mediators of the pandemic's mental-health impact, extending the perceived-control and psychological-distance account of Zheng, Miao and Gan (2020). These frameworks imply that local case counts matter partly through how individuals interpret them, not only through actual infection. They also provide a coherent explanation for adaptation: beliefs, expectations and coping strategies can change even when objective exposure remains high. Psychological mechanisms therefore connect the early cross-sectional evidence with later findings that mean life satisfaction often recovered before all pandemic conditions had disappeared.

Behavioural and economic research identifies more directly observable pathways. Han et al. (2022) and Zheng et al. (2022) documented pandemic-related changes in nutritional intake and dietary consumption in rural and provincial China. Dui (2022) found a widening gender wage gap during the pandemic, while Wan, Wang and Jiang (2023) reported

greater regional income inequality across 20 Chinese provinces. These studies show that disease and restrictions can affect life satisfaction through material living conditions as well as psychological distress. Wang et al. (2026) linked educational convergence to the equalisation of happiness in China, and Ding et al. (2025) incorporated psychological burdens into a broader post-pandemic research agenda. These mechanisms should not be treated as separate chains: income loss can constrain consumption and increase anxiety, while psychological resources can shape an individual's capacity to absorb economic disruption.

2.4 Heterogeneity in Pandemic Effects

2.4.1 Older Adults, Isolation and Resilience

Evidence on older adults resists a simple vulnerability narrative. Di Gessa and Zaninotto (2023) examined changes in life satisfaction before and during the pandemic among older people in England, while Polinesi, Ciommi and Gigliarano (2024) examined multidimensional well-being, including social connections, among older Europeans. Both place social connectedness at the centre of age-related responses, but neither implies that chronological age alone determines the outcome. Schlomann et al. (2022) found substantial psychosocial adaptation among older Germans, and Kung and Steptoe (2024) similarly showed that average resilience could coexist with adverse outcomes among socially isolated individuals. Lalani et al. (2023) supplied an explanatory bridge by modelling coping style as a mediator of life satisfaction among older adults. Read together, these findings resolve an apparent contradiction: older people may face greater health risks yet retain routines, expectations or coping resources that protect evaluative wellbeing, whereas larger losses are concentrated where isolation or weak coping removes those protections. Age is therefore better treated as a source of effect modification than as a sufficient explanation in itself.

2.4.2 Younger Adults and Other Social Divisions

Findings for younger people are more consistently adverse. Henseke, Green and Schoon (2022) associated the second pandemic phase with changes in young people's subjective wellbeing, and Li and Gutman (2024) and Larson et al. (2023) showed how pandemic disruption intersected with the transition to adulthood. Education, entry into employment, independent living and relationship formation are time-sensitive processes; delaying them

may have consequences that contemporaneous infection or lockdown measures do not fully capture. The contrast with older adults therefore does not imply that disease was objectively more severe for the young. It may instead reflect the greater life-course value of opportunities displaced by restrictions. This distinction matters for empirical interpretation: an age interaction can identify differences in responses to provincial exposure, but it cannot by itself reveal whether those differences arise from employment, education, household position or social contact. Age-specific estimates should therefore be presented as evidence of heterogeneity rather than as direct tests of any single life-course mechanism.

Other social divisions reinforce the same principle. Benavente et al. (2025) found distinct trajectories of psychological distress and life satisfaction for migrants and non-migrants in Norway, and Moreno-Agostino, Das-Munshi and Ploubidis (2023) traced changes in sex differences across long-term life-satisfaction trajectories. Dui (2022) documented a widening gender wage gap in China, providing a material route through which demographic inequalities could enter wellbeing. These studies caution against searching for a single universally vulnerable category. Exposure and resources are distributed across several overlapping dimensions, so an observed regional or demographic difference may reflect their combination. For the present study, age and region are substantively justified because the research questions pre-specify them and the available panel supports their analysis. Other covariates can help identify groups that experienced larger declines, but exploratory subgroup results require restraint: multiple comparisons raise the probability of chance findings, and a statistically different coefficient does not establish the mechanism that produced it. The literature therefore motivates pre-specified heterogeneity tests, transparent reporting of uncertainty and a clear distinction between confirmatory and exploratory evidence.

2.5 Methodological Limitations and Research Gap

2.5.1 Measurement and Outcome Comparability

The reviewed evidence is not fully comparable because studies operationalise both exposure and wellbeing differently. Case counts, deaths, self-reported infection and perceived risk represent distinct dimensions of disease severity, just as binary lockdown indicators, policy phases and continuous stringency measures capture different aspects of restriction. Outcomes range from life satisfaction to psychological distress, loneliness and

general mental health. These measures are related but not interchangeable: life satisfaction is a cognitive evaluation, whereas distress and loneliness capture more specific experiences. Apparent disagreement across studies may therefore reflect measurement choices rather than genuinely different effects. Official case data also depend on testing and reporting, while policy indicators may record formal rules without capturing implementation or compliance. Mao, Pepermans and Beutels (2025) illustrate the value of cross-country comparison, but such designs necessarily average across institutional and cultural differences. A subnational design improves contextual comparability, although province-level exposure remains a proxy for the experience of individuals within each province.

2.5.2 Endogeneity and Causal Interpretation

Identification remains the more fundamental limitation. Policy responds to disease, behaviour responds to policy, and prior wellbeing may influence compliance; time-varying economic conditions can also affect both government decisions and life satisfaction. Individual fixed effects remove stable personal characteristics, and region-by-wave fixed effects absorb broad time shocks that vary across large regions, but neither adjustment guarantees that province-specific, time-varying confounding has disappeared. Continuous difference-in-differences is useful when every province is exposed at a different intensity and no genuinely untreated province exists. Its coefficients nevertheless rely on a parallel-trends-style assumption: in the absence of differential exposure, changes in life satisfaction would not have been systematically related to the dose. Where exact interview dates are unavailable, province-wave exposure also cannot establish fine temporal ordering within a survey year. The design can therefore provide stronger conditional dose-response evidence than a cross-sectional regression, but causal language must remain proportionate to the available timing information and the strength of the robustness tests.

2.5.3 The Joint-Exposure and Subnational Gap

Within the corpus reviewed here, three gaps remain. First, disease severity and containment stringency are usually studied separately even though the literature shows that they are conceptually distinct and empirically interdependent. Ma, Wang and Wu (2024) provide the closest China-based precedent by treating lockdown and infection as separable time-varying exposures, while Grimes (2022) explicitly calls for the impacts of the pandemic and lockdown to be measured apart. Second, much of the international

evidence is cross-national, whereas China-focused studies more often examine mental health, income, consumption or inequality than province-level life-satisfaction responses to both exposures. Third, heterogeneity is frequently documented *ex post* rather than integrated into the main exposure design. The gap is therefore not an absence of all relevant components, but the limited integration of continuous subnational disease burden, policy stringency, individual panel change and pre-specified heterogeneity within a single framework. This narrower formulation avoids overstating novelty and identifies more precisely what the present analysis can contribute.

A further gap concerns interpretation rather than model specification. Including both exposures in one equation estimates the association of each conditional on the other; it does not make them experimentally independent. The disease coefficient may partly absorb unmeasured policy implementation, while the policy coefficient may incorporate behavioural responses and perceived risk. Nor does a larger raw coefficient necessarily imply that one shock is substantively more important, because the two variables differ in scale and distribution. Meaningful comparison therefore requires standardised changes, common outcome units or clearly defined counterfactual contrasts. Phase-specific estimates introduce an additional interpretive problem: differences between 2020 and 2022 may reflect adaptation, altered policy implementation, vaccination, reporting changes or a different composition of risk. The literature supports testing whether effects vary across phases, but it also requires the discussion to distinguish what the estimates directly demonstrate from mechanisms that remain plausible explanations.

2.6 Positioning of the Present Study

The present study responds to this integrated gap using the China Family Panel Studies from 2010 to 2022. It links repeated individual observations to province-wave measures of pandemic exposure and implements a continuous difference-in-differences specification with individual fixed effects and region-by-wave fixed effects. Disease severity is represented by logged cumulative confirmed cases, while policy stringency is represented by the number of months under strict containment. The analysis first estimates each exposure separately and then jointly, making changes in the coefficients visible rather than obscuring them within a single specification. It also compares pandemic phases and evaluates heterogeneity by age and region, with other covariates used cautiously for exploratory subgroup analysis. This design extends the trajectory and panel literature,

including Molla et al. (2026), and develops the subnational China evidence represented by Ma, Wang and Wu (2024) and Wan, Wang and Jiang (2023).

The intended contribution is therefore comparative and methodological rather than a claim to have fully separated two exogenous treatments. The study asks whether within-person changes in life satisfaction covary more strongly with province-level disease burden or with containment duration after common observed structure is controlled, whether those relationships differ across pandemic phases, and whether they are modified by age or region. The long pre-pandemic panel helps establish baseline trajectories, while fixed effects reduce bias from stable individual differences and broad regional shocks. At the same time, province-wave measurement, the absence of exact interview dates and the mutual determination of disease and policy bound the causal interpretation. Stating these limits does not weaken the study's purpose; it defines the scope of a credible contribution. The analysis integrates exposures that the literature often separates, tests the four research questions within a common framework and reports the resulting estimates as conditional continuous-treatment effects supported by explicit assumptions and robustness checks.

3 Methodology

3.1 Research Questions

This study examines how provincial COVID-19 disease burden and containment-policy stringency were associated with within-person changes in life satisfaction among Chinese residents. Four questions organise the empirical analysis:

- (1) How was provincial COVID-19 disease severity associated with individual life satisfaction in China, conditional on measured containment-policy stringency?
- (2) How was provincial containment-policy stringency associated with individual life satisfaction in China, conditional on measured disease severity?
- (3) Did these conditional associations differ in magnitude on a standardised exposure scale, and did each association differ between the 2020 and 2022 survey waves?
- (4) Did these associations vary across age groups and regions?

All provinces experienced the pandemic, leaving no genuinely untreated province for a conventional treated–control comparison. I therefore use a difference-in-differences framework with continuous exposure intensity. Conditional on individual and region-by-wave fixed effects, identification relies on differences in provincial exposure intensity within the same region and survey wave. The estimates describe conditional relationships; stronger causal interpretations require the assumptions discussed below.

3.2 Data and Sample

3.2.1 Data Sources

The study combines seven waves of the China Family Panel Studies (CFPS), conducted in 2010, 2012, 2014, 2016, 2018, 2020, and 2022, with province-level measures of COVID-19 cases and containment policies. The longitudinal structure follows respondents across waves, allowing individual fixed effects to absorb stable personal characteristics and persistent response tendencies.

Policy exposure is drawn from the subnational files of the Oxford COVID-19 Government Response Tracker (OxCGRT). The Chinese records used here identify policies at the provincial rather than city level, so both pandemic exposures are assigned at the province-by-wave level. The exposure windows are 1 January–30 June 2020 and 1 January–30

April 2022; both variables are coded as zero in the five pre-pandemic waves. Exact interview dates are unavailable in the CFPS files used here. The windows therefore represent wave-level pandemic context rather than respondent-specific exposure histories, and some measured exposure may postdate an interview. Temporal ordering cannot be established for every respondent, which limits causal interpretation.

3.2.2 Sample Construction

The main analytical sample contains 17,733 person-wave observations from 3,825 adults across 25 provinces. Respondents are retained if they have valid life-satisfaction and province-of-residence information and appear in at least two waves. Individuals who move across provinces are excluded to maintain consistent exposure assignment. Provinces with insufficient observations and records missing variables required for a given specification are also excluded. The panel is unbalanced, so wave-level descriptive means may reflect changes in sample composition as well as within-person change.

3.2.3 Variable Construction

3.2.3.1 Life Satisfaction

The outcome is self-reported life satisfaction on a five-point scale, with higher values indicating greater satisfaction. The analysis treats the scale as approximately continuous and estimates linear models with individual fixed effects. Coefficients are therefore interpreted in life-satisfaction scale points. This provides a transparent interpretation across specifications but assumes approximately equal distances between adjacent response categories. The estimates remain specific to this linear treatment of an ordinal outcome; they do not establish robustness to alternative outcome coding or ordinal estimators.

3.2.3.2 Disease Severity

Disease severity is defined as the natural logarithm of one plus the mean cumulative confirmed-case rate per 100,000 residents within each exposure window. For province p and wave t , the measure is written as:

$$Disease_{pt} = \ln(1 + MeanCumCaseRate_{pt})$$

This cumulative measure is intended to capture the scale of pandemic experience accumulated by each survey wave, including potential fear, bereavement, and social disruption. The logarithmic transformation reduces the leverage of extreme provinces while

allowing the marginal relationship to diminish as disease burden rises. A one-unit increase in the transformed measure multiplies one plus the underlying case rate by e ; substantive interpretation is therefore supplemented with contrasts across observed percentiles. Because confirmed cases depend on testing capacity, reporting practice, and case definitions, robustness checks replace the preferred logged cumulative measure with the untransformed cumulative case rate and with logged and untransformed new-case rates. These alternative definitions distinguish sensitivity to functional form from sensitivity to whether exposure captures accumulated burden or recent incidence. Results are reported in Appendix C.

3.2.3.3 Containment-Policy Stringency

Containment intensity is measured as the number of days on which the provincial Stringency Index (SI) is at least 70, divided by 30. It is expressed approximately in months of high-stringency exposure:

$$Lockdown_{pt} = \frac{HighStringencyDays_{pt}}{30}$$

The threshold of 70 is a researcher-defined operational cut-off, not an official OxCGRT definition of lockdown. Although the variable is labelled Lockdown in the equations, it measures the duration of high composite policy stringency. It does not directly capture enforcement, compliance, neighbourhood closures, or the policy mix experienced by each respondent. The reported policy estimates concern this duration measure; their interpretation does not extend automatically to other thresholds, the mean SI, or individual policy components. The principal policy measure is the number of months for which the provincial Stringency Index is at least 70. Because this threshold is researcher-defined, the resulting coefficient is interpreted specifically as the association with the duration of high composite stringency. It should not be assumed to generalise to alternative thresholds, average stringency, or individual policy components. Sensitivity to specification choices is examined separately in Appendix C where the available alternative policy estimates permit.

3.3 Research Design

3.3.1 Continuous-Intensity Difference-in-Differences Design

All provinces were exposed to the pandemic, so there is no genuinely untreated group after 2020. The preferred design is therefore a continuous-intensity difference-in-

differences framework that asks whether life satisfaction changed more in provinces exposed to greater pandemic intensity. The baseline joint specification is:

$$LS_{it} = \beta_D Disease_{pt} + \beta_L Lockdown_{pt} + \alpha_i + \lambda_{rt} + \varepsilon_{it}$$

where LS_{it} denotes life satisfaction for individual i in wave t ; $Disease_{pt}$ and $Lockdown_{pt}$ are the two province-level exposures; α_i denotes individual fixed effects; and λ_{rt} denotes region-by-wave fixed effects for eastern, central, western, and north-eastern China. Among respondents who do not move across provinces, individual fixed effects remove all stable individual characteristics and time-invariant provincial differences. Region-by-wave fixed effects allow the four broad regions to follow different time paths. Conditional on these effects, identification comes from differences in exposure intensity across provinces within the same region and survey wave.

3.3.2 Interpretation and Comparison of Exposure Effects

In the joint model, β_D measures the life-satisfaction difference associated with a one-unit increase in logged cumulative disease burden, conditional on measured policy stringency and the other model terms. β_L measures the corresponding difference for one additional month of high-stringency exposure, conditional on measured disease burden and the other terms. These are conditional dose–response parameters. Policy responds to cases and may also affect subsequent cases, so mutual adjustment can block part of a policy pathway or introduce simultaneity bias. The joint regression does not separately identify the total causal effect of each channel, and a joint test of the exposure coefficients does not measure the total pandemic effect.

To compare magnitudes, the exposure coefficients are expressed per one-standard-deviation increase in each exposure and their difference is tested directly. Separate significance tests cannot establish that one association is larger than the other. To assess temporal variation, wave-specific models interact the exposures with indicators for 2020 and 2022 and test equality of the two wave-specific coefficients for each exposure. These comparisons concern survey-wave differences, which may also reflect changes in exposure windows, reporting practices, and sample composition; they do not isolate the effects of particular viral variants or policy regimes.

3.3.3 Covariate Selection

The baseline specification deliberately avoids covariates that may themselves be affected by the pandemic. Self-rated health, employment, income, marital status, and insurance participation are plausible mediators; conditioning on their contemporaneous values would change the estimand from a total exposure effect to a more restricted conditional relationship. The same concern applies to provincial GDP, consumption, and industrial activity measured in 2020 or 2022. The parsimonious fixed-effects model is therefore reported first. Predetermined characteristics may enter through 2018 baseline values interacted with wave indicators, or through counterfactual provincial trends estimated exclusively from pre-2020 data. Models that add contemporaneous controls or hold provincial characteristics at their 2018 levels are treated as sensitivity analyses. Freezing a provincial value at its 2018 level constitutes baseline adjustment; it does not generate a data-driven province-specific trend.

3.4 Identification Assumptions and Statistical Inference

3.4.1 Parallel-Trends Assumption

A causal interpretation requires conditional parallel trends: in the absence of the measured pandemic exposures, provinces within the same broad region that later received different exposure doses would have experienced comparable changes in life satisfaction. The design additionally requires no anticipation, no unmeasured province-by-wave shocks correlated with exposure intensity, stable measurement of the exposure proxies, and no consequential cross-province spillovers. Pre-trends are assessed by interacting pre-pandemic wave indicators with a fixed post-pandemic exposure measure and jointly testing the pre-pandemic interaction coefficients. Event-study estimates are reported with confidence intervals and evaluated by their magnitude and pattern rather than by p-values alone. Failure to reject the joint pre-trend test does not prove parallel trends, particularly with only 25 provinces. Key alternative specifications and sensitivity analyses are reported transparently in Appendix C, while the preferred specification is justified on substantive and design grounds rather than selected solely according to post-pandemic statistical significance.

3.4.2 Province-Level Clustered Inference

Both exposures vary at the province-by-wave level, so standard errors are clustered by province. With 25 clusters, conventional cluster-robust inference may be unreliable in finite

samples. The reported main tests are therefore supplemented with a province-level wild cluster bootstrap imposed under the null, with bootstrap p-values also available for selected subgroup and wave-specific estimates. Comparisons without a reported bootstrap p-value are interpreted on the basis of their conventional clustered inference alone. Clustering by city would not increase the number of independent exposure units because cities within a province share the assigned policy measure.

Statistical precision is assessed using confidence intervals as well as p-values. These are approximate descriptions of precision under the fitted design, not evidence that smaller associations are absent or that the observed estimate has 80% power. A statistically imprecise coefficient is interpreted through the range of values compatible with its confidence interval, rather than as proof of no relationship.

3.4.3 Robustness and Sensitivity Checks

Robustness analyses examine whether the principal findings depend on exposure measurement, covariate adjustment, influential provinces, or selected survey waves. For disease severity, the preferred logged cumulative measure is compared with the raw cumulative rate and with logged and raw new-case rates. Alternative covariate specifications vary individual and provincial adjustment, including a specification using observed post-2020 macroeconomic values. Influence is assessed through leave-one-province-out re-estimation, while an additional specification excludes the 2018 wave. A within-region permutation exercise is used only as a placebo diagnostic because provincial exposure cannot be regarded as randomly assigned within regions. Detailed estimates are reported in Appendix C.

3.5 Heterogeneity Analysis

Age and region are the focal dimensions of heterogeneity because they directly address Research Question 4. Age is measured at the 2018 baseline and interacted with each exposure. A continuous interaction assesses a linear age gradient; supplementary groups of 44 or younger, 45–59, and 60 or older allow the pattern to depart from linearity. The baseline age main effect is absorbed by individual fixed effects. Grouped estimates are assessed using tests of equality across slopes, so differences in within-group significance are not treated as evidence of between-group differences.

Exploratory comparisons distinguish agricultural from non-agricultural hukou, urban from rural residence, and education below lower secondary from lower secondary or above.

Hukou and actual residence capture different dimensions and are not interchangeable. Current residence is potentially responsive to pandemic conditions, so residence-based comparisons cannot be assumed to identify differences between predetermined groups. The hukou and education contrasts likewise do not, by themselves, establish that group membership was fixed before the pandemic. These comparisons describe differences in conditional associations and do not identify the employment, insurance, or mobility mechanisms that might explain them.

Regional heterogeneity is assessed by allowing the exposure slopes to vary across eastern, central, western, and north-eastern China and testing whether the resulting coefficients are equal across regions. These estimates are interpreted cautiously because the number of independent provincial exposure units is small within some regions, particularly the north-east. Regional subgroup coefficients are therefore used to assess possible heterogeneity rather than to establish precise region-specific causal effects.

The reported subgroup tests are not adjusted for multiple testing. Their nominal p-values therefore require caution, including for the focal age and regional comparisons. Direct interaction or between-group tests, where reported, are distinguished from subgroup-specific tests; a significant coefficient in one group and a non-significant coefficient in another is insufficient to establish heterogeneity. The subgroup findings are interpreted as exploratory evidence rather than a definitive ranking of vulnerability.

3.6 Methodological Limitations

Several limitations qualify the estimates. Province-level exposure masks local variation, and recorded cases depend on testing and reporting as well as underlying disease burden. Missing interview dates prevent precise exposure–outcome ordering. Survey-mode changes and attrition may affect sample composition and responses. The estimates also depend on the exposure definition, covariate adjustment, and specification choices, while 25 provincial clusters constrain precision. Reciprocal determination of cases and policy prevents the joint model from recovering two independent total causal effects, and unadjusted subgroup tests leave a risk of chance findings. The analysis is therefore presented as panel evidence on conditional associations, with causal interpretation contingent on assumptions that the reported diagnostics cannot establish.

4 Results and Discussion

This section addresses the four research questions through a sequence of descriptive evidence, identification diagnostics, main estimates and heterogeneity analyses. The analysis distinguishes provincial disease severity from containment policy stringency and examines how each exposure is associated with within-person changes in life satisfaction. Coefficients are reported in points on the original one-to-five life satisfaction scale. Given the identification limits discussed in the methodology, I interpret the estimates as conditional associations; a causal interpretation would require additional assumptions that the available diagnostics cannot establish.

4.1 Descriptive Evidence

4.1.1 Sample Sources

The final estimation sample contains 17,733 person-wave observations from 3,825 respondents in 25 provinces across seven CFPS waves from 2010 to 2022. This panel structure allows the analysis to use changes within the same respondent rather than relying on comparisons between different people observed at different times. In the descriptive file, unadjusted mean life satisfaction is 4.001 in 2018, 4.023 in 2020, and 3.972 in 2022. The national sample mean therefore changes little across the pandemic period and remains far more stable than in some earlier intervals, such as the increase of about 0.47 points between 2012 and 2014. Viewed only at this aggregate level, the series gives little indication of a large common decline after 2018.

A stable national sample mean can conceal substantial differences in exposure. Provinces experienced very different disease burdens and policy histories, so declines in highly exposed areas may be offset by stable or improving outcomes elsewhere. Changes in wave composition can also enter an unadjusted national average because the number and identity of respondents vary across surveys. The descriptive series therefore provides context for the pandemic-period pattern but does not isolate a change attributable to the pandemic. The main empirical question is whether the same individuals experienced systematically different changes when their provinces faced different levels of disease burden and containment.

The continuous-exposure design requires meaningful variation in both exposures, and the final data provide it. The logged cumulative disease measure ranges from 0.313 in Guizhou to 4.655 in Hubei, with Shanghai and Jilin also near the upper end of the

distribution. Strict containment duration ranges from 0.17 months in Chongqing to 3.78 months in Heilongjiang. The two exposure measures do not rank provinces in the same way. Hubei combines exceptionally high cumulative disease burden with about two months of strict containment, while Heilongjiang records a more moderate disease burden but the longest strict containment duration. Shanghai also has a high disease exposure without ranking equally high on containment duration. This separation provides empirical variation for jointly estimating the conditional associations of the two exposure measures.

Variation alone does not establish exogeneity. Provinces with larger populations, greater mobility, or stronger economic activity could report more cases, while policy stringency often responded to the outbreaks reflected in the disease measure. The exposure distribution therefore reveals both the variation used for estimation and its main limitation. Fixed effects leave usable within-region variation, but that variation may still correlate with province-specific changes in economic conditions, survey implementation, or public expectations. I interpret the multivariate estimates considering this constraint and do not assume that provincial exposure was randomly assigned.

4.1.2 Parallel Trends Assumption Test

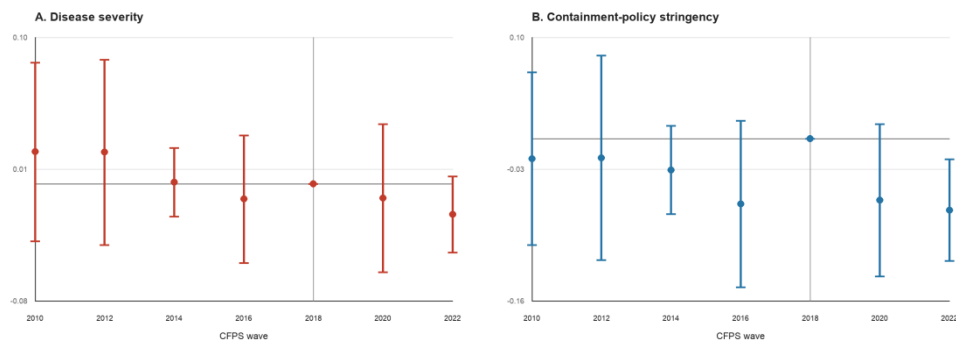


Figure 1 Parallel Trends Assumption Test

Figure 1 reports the event-study estimates from the preferred specification, with 2018 as the reference wave. For disease severity, the coefficients for 2010, 2012, 2014, and 2016 are 0.0218, 0.0215, 0.0010, and -0.0105. The joint test gives $F = 0.927$ and $p = 0.465$, while the pre-period linear slope test gives $p = 0.650$. For containment stringency, the corresponding coefficients are -0.0197, -0.0189, -0.0310, and -0.0643. Their joint test gives $F = 0.956$ and $p = 0.449$, and the slope test gives $p = 0.770$. Neither set of tests detects statistically significant pre-pandemic differences at conventional levels, although the confidence intervals remain wide.

The joint null hypothesis examined for each exposure can be written as

$$H_0: \theta_{2010} = \theta_{2012} = \theta_{2014} = \theta_{2016} = 0.$$

These diagnostics are compatible with conditional parallel trends, but they do not establish the assumption. Only 25 provinces contribute independent exposure variation, which limits power to detect modest pre-existing differences. The preferred specification was also chosen after comparing alternative time windows and fixed-effect structures. Region-by-wave fixed effects have a substantive rationale because the four broad regions followed different long-run trajectories, particularly in the north east, but model comparison still creates a risk of specification searching. I therefore interpret the event-study diagnostics as failing to reject the pre-trend restrictions in the preferred model. A p value above 0.05 does not establish that subsequent differences can be attributed to the measured exposure.

4.2 Research Question 1: Disease Severity and Life Satisfaction

4.2.1 Average Estimate

The preferred joint model shows a negative association between provincial disease severity and life satisfaction. The coefficient on logged cumulative case burden is -0.0331 , with a province-clustered standard error of 0.0117 and a 95% confidence interval from -0.0572 to -0.0090 . The conventional clustered test gives $p = 0.009$, and the null-imposed wild cluster bootstrap gives $p = 0.019$. Conditional on the model's covariates and fixed effects, a one-unit increase in logged cumulative disease burden is associated with a 0.033-point lower life satisfaction score, holding measured containment duration constant. The estimated association is modest on a one-to-five scale. The reported clustered confidence interval excludes zero, and both reported p values are below 0.05; these inference results do not by themselves establish causality.

Table 1 Average conditional associations of provincial disease severity and strict containment duration

Exposure	Est.	SE	95% CI	Clust. p	Wild p	MDE
Logged cumulative disease burden	-0.0331	0.0117	[-0.0572, -0.0090]	0.009	0.019	0.0341
Strict containment months	-0.0172	0.0135	[-0.0450, 0.0106]	0.213	0.194	0.0393

The two pandemic exposures are jointly associated with life satisfaction in the preferred model, with $F = 10.393$ and $p = 0.001$. This test rejects the restriction that both conditional coefficients equal zero, but it does not estimate the total effect of the pandemic. National consequences shared within a region and survey wave are absorbed by the fixed effects, while the reported coefficients use differential provincial exposure. The result therefore indicates that the two exposure measures jointly explain part of the within person variation that remains after adjustment. The stronger precision of the disease coefficient and the wider uncertainty around the policy coefficient point to unequal evidential strength rather than a complete decomposition of the pandemic into two causal components.

4.2.2 Magnitude Interpretation

Standardisation places the disease estimate on a more interpretable scale. One standard deviation of logged cumulative burden equals 0.738 units in the estimation sample and corresponds to a 0.0244 point decline in life satisfaction. Relative to the 1.03 point standard deviation of the outcome, the average association is about 0.024 outcome standard deviations. The average slope is small, but it is a marginal dose response estimate rather than the change from no pandemic to the observed pandemic. Provincial exposure also spans a wide range. Applying the common coefficients to observed provincial doses gives a combined model implied change of -0.188 points for Hubei, including -0.154 points from the disease component. This calculation is descriptive and extrapolates toward zero exposure, yet it shows how a modest average slope can generate larger differences at the upper end of the exposure distribution.

Several mechanisms could be consistent with this negative association. Cumulative disease burden may be linked to higher perceived mortality risk, illness and bereavement within social networks, pressure on local health services, and prolonged uncertainty around ordinary activity. Life satisfaction measures a broad evaluation of life rather than a clinical symptom, so the coefficient may capture associations with several forms of disruption at once. The regression does not identify which pathway dominates and cannot separate actual infection from information about local outbreaks. I therefore interpret the estimate at the level of provincial disease burden, without attributing a known causal change to confirmed infection for each respondent.

4.2.3 Sensitivity and Limits

The disease result depends on how exposure is measured. The preferred logged cumulative measure gives -0.0331 with $p = 0.009$. The raw cumulative rate also produces a negative estimate, whereas logged and raw new-case measures are smaller and statistically imprecise. On the standardised exposure scale, the estimated associations are -0.0244 for logged cumulative cases, -0.0123 for the raw cumulative rate, -0.0111 for logged new cases, and -0.0072 for raw new cases. The evidence is strongest when exposure represents accumulated pandemic experience. That pattern is compatible with an association between life satisfaction and an extended history of disruption, but it does not establish this mechanism. It also defines an important boundary of the finding because the strength of the evidence changes with the exposure concept.

Covariate choice is associated with a larger shift in the estimate. Without additional controls, the disease coefficient is -0.0132 with $p = 0.336$, and the model with only individual covariates gives -0.0143 with $p = 0.325$. Adding the pre-pandemic provincial adjustment changes the estimate to -0.0314 with $p = 0.007$, while the full model gives -0.0331 with $p = 0.009$. One interpretation is negative confounding if economically developed provinces combined higher reported case burdens with more favourable satisfaction trajectories. The same pattern also shows that the main finding depends on a particular adjustment strategy. Fixing provincial covariates at their 2018 values does not reconstruct province specific counterfactual trends. I therefore treat the preferred estimate as evidence consistent with a negative disease relationship under this adjustment rather than as an unconditional causal estimate.

4.3 Research Question 2: Containment Policy Stringency and Life Satisfaction

4.3.1 Average Estimate and Statistical Precision

The joint model does not detect a statistically precise average association between strict containment duration and life satisfaction. One additional month with a provincial Stringency Index at or above 70 is associated with a -0.0172 -point difference in life satisfaction, conditional on measured disease burden, covariates and fixed effects. The province-clustered standard error is 0.0135, and the 95% confidence interval ranges from -0.0450 to 0.0106. The conventional test gives $p = 0.213$, while the wild cluster bootstrap gives $p = 0.194$. The point estimate is negative, but the interval includes zero, moderately

negative values and small positive values. The evidence therefore does not establish a statistically detectable average association, nor does it establish the absence of an association.

The minimum detectable effect helps explain this uncertainty. At 80% power and a 5% significance level, the design can reliably detect an effect of about 0.0393 satisfaction points per additional strict containment month. The observed estimate is less than half that magnitude. Precision is constrained because policy exposure varies across only 25 provinces, while region-by-wave fixed effects remove substantial between-region variation. Treating cities as independent policy units would not solve this problem because the policy measure does not vary independently within provinces. The confidence interval indicates the range of average conditional associations compatible with the model and data; smaller associations remain unresolved, including those that could correspond to larger model-implied differences over several months.

4.3.2 Interpretation of the Average Null Result

A small average coefficient can coexist with different associations for particular groups. Restrictions may be associated with lower satisfaction through work disruption, mobility constraints, weaker social contact, and reduced access to services. They may also be associated with lower perceived infection risk and a sense of public protection. The balance between these possible costs and benefits may vary with employment, household resources, age, and pandemic stage, so differing responses could be compatible with an average coefficient near zero. The later heterogeneity results show more negative containment estimates among respondents with nonagricultural hukou and those with at least lower secondary education, while the estimate for agricultural hukou holders is close to zero. These exploratory patterns are consistent with subgroup differences, but they do not demonstrate that offsetting mechanisms account for the average result.

Exposure measurement can also attenuate the estimate. The strict containment variable counts the duration for which a composite provincial index exceeds a researcher defined threshold. It does not measure enforcement, compliance, neighbourhood closure, or the exact policy mix experienced before each interview. Provinces with the same number of high stringency days may differ in workplace closure, stay at home requirements, service provision, and local implementation. This measurement gap can weaken the estimated slope. The average null result therefore applies to the provincial duration measure used

here and should not be generalised to every form of containment or to each respondent's actual experience of restrictions.

4.4 Research Question 3: Relative Magnitude and Differences Between Survey Waves

4.4.1 Comparing the Two Exposure Channels

The results do not establish a statistical ranking of the two exposure associations by magnitude. Because logged disease burden and strict containment duration use different units, I compare coefficients after standardising both exposures. A one-standard-deviation increase in disease severity is associated with -0.0244 satisfaction points, while the corresponding containment estimate is -0.0154 points. Their difference is -0.0091 , with a standard error of 0.0184 and $p = 0.627$.

$$\Delta\beta_{std} = \beta_{D,std} - \beta_{L,std} = -0.0091.$$

The disease coefficient is more negative and individually more precise, but that pattern does not establish that the underlying disease association is larger in magnitude. Separate significance tests do not substitute for a direct test of the coefficient difference.

Table 2 Standardised associations of disease severity and containment stringency with life satisfaction

Term	Est.	SE	Clust. p	Wild p
Disease severity, standardised	-0.0244	0.0086	0.009	NA
Stringency, standardised	-0.0154	0.0120	0.213	NA
Difference, disease minus stringency	-0.0091	0.0184	0.627	NA
Disease interaction, 2020	-0.0577	0.0228	0.018	0.045
Disease interaction, 2022	-0.0262	0.0105	0.020	0.023
Stringency interaction, 2020	-0.0049	0.0192	0.799	NA
Stringency interaction, 2022	-0.0344	0.0247	0.177	NA

The more defensible distinction concerns precision rather than rank ordering. The plausible ranges of the two standardised coefficients overlap substantially, while the disease estimate is more precisely measured in the preferred specification. This asymmetry could reflect a more consistent association between disease burden and wellbeing, but it could also arise because the disease measure retains more usable within-region variation. Containment exposure has a narrower residual distribution after fixed

effects and may contain more measurement error. The results provide stronger evidence for a negative disease association than for a negative average policy association, while leaving the relative magnitude of the underlying conditional associations unresolved.

4.4.2 Differences Between the 2020 and 2022 Waves

The disease relationship is more negative in 2020 than in 2022. The wave specific coefficient is -0.0577 in 2020, with $SE = 0.0228$, $p = 0.018$, and wild $p = 0.045$. In 2022 it is -0.0262 , with $SE = 0.0105$, $p = 0.020$, and wild $p = 0.023$. The 2020 point estimate is more than twice the 2022 estimate in absolute value, but the formal difference is only weakly distinguished from zero, with a difference of -0.0315 , $SE = 0.0184$, and $p = 0.100$. The pattern is consistent with a stronger disease related shock during the acute phase and a smaller relationship that persists in 2022. The evidence does not support a precise claim about the proportion by which the effect declined.

Containment shows the opposite point estimate pattern. The coefficient is -0.0049 in 2020, with $SE = 0.0192$ and $p = 0.799$, and -0.0344 in 2022, with $SE = 0.0247$ and $p = 0.177$. Neither wave specific coefficient is statistically distinguishable from zero, and their difference is also imprecise with $p = 0.394$. The estimates are compatible with the possibility that repeated or prolonged restrictions became more costly for subjective wellbeing by 2022, but the data do not confirm that mechanism. The significant 2022 event study coefficient for stringency, -0.0706 with $p = 0.011$, comes from a different specification and answers a different question. It does not replace the non significant formal comparison between the two post pandemic waves.

4.4.3 Interpretation of Temporal Variation

The temporal pattern has several plausible explanations. In 2020, the disease represented a new and poorly understood health threat, with unusually high uncertainty around mortality, transmission, and health-system capacity. Emergency restrictions may also have been perceived as temporary collective protection, which could have moderated their association with life satisfaction. By 2022, knowledge of the virus had improved, while repeated restrictions, employment disruption, and uncertainty around reopening had accumulated. These contextual explanations are consistent with a less negative disease coefficient and a more negative containment point estimate, but they are not directly tested. The two survey waves also differ in exposure-window length, circulating variants, testing practice, survey mode, and sample composition. The interactions therefore

describe wave-specific associations and do not isolate the contribution of a particular viral variant or policy regime.

4.5 Research Question 4: Heterogeneity Across Population Groups and Regions

4.5.1 Age

Age heterogeneity is assessed with age fixed at the 2018 baseline. In the continuous interaction model, the disease-by-age coefficient is 0.0008, with $SE = 0.0008$, $p = 0.333$, and wild $p = 0.286$. This result provides little evidence of a linear age gradient in the disease association. The containment-by-age interaction is 0.0011, with $SE = 0.0006$, clustered $p = 0.055$, and wild $p = 0.038$. Given the negative main containment slope, the positive interaction corresponds to an estimated association that is about 0.011 points less negative for every additional ten years of baseline age. The wild bootstrap result is compatible with an age gradient at the 5% level, while the conventional clustered result does not reach that threshold. This difference between inference procedures, together with the wider set of heterogeneity tests, warrants an exploratory interpretation.

Grouped estimates show a less linear pattern. For disease severity, the coefficient is -0.0563 among respondents aged 44 or younger at baseline, with $SE = 0.0148$, $p = 0.001$, and wild $p = 0.007$. The estimate is -0.0260 among those aged 45 to 59, with $p = 0.341$, and -0.0409 among those aged 60 or older, with $p = 0.126$ and wild $p = 0.080$. A joint test rejects equality across the three disease coefficients, with $F = 6.279$ and $p = 0.003$, before adjustment for multiple testing. The reported estimates therefore suggest age-group heterogeneity without a simple monotonic gradient. The youngest group has the most negative and most precise estimate, while the oldest group also has a negative point estimate with much greater uncertainty.

Younger adults may have experienced a broader economic and social disturbance. Employment entry, job mobility, income growth, partnership formation, and social contact outside the household are especially salient at younger ages and may be associated with local outbreaks even without direct infection. Older adults may have faced greater direct health risk, while retirement income and a smaller pre-pandemic activity radius could have limited disruption in other domains. The data do not test these mechanisms directly. The finding concerns differences in the estimated life satisfaction association and should not be

read as evidence that the health consequences of COVID-19 were less severe for older adults.

4.5.2 Hukou Status and Place of Residence

Hukou status shows one of the clearest reported subgroup contrasts in the containment association. The containment coefficient is -0.0565 for respondents with nonagricultural hukou, with $SE = 0.0151$, $p = 0.001$, and wild $p = 0.003$. For agricultural hukou holders, the estimate is -0.0024 , with $SE = 0.0157$, $p = 0.882$, and wild $p = 0.865$. The difference between the two slopes is -0.0541 , with $SE = 0.0171$ and $p = 0.004$. Disease severity does not show the same split. Its coefficient is -0.0368 among agricultural hukou holders and -0.0208 among nonagricultural hukou holders, while their difference is not statistically precise with $p = 0.376$. Before multiple-testing adjustment, the evidence therefore suggests hukou heterogeneity in the containment association, without establishing a general hukou difference across both exposures.

Table 3 Selected heterogeneity estimates for the relationships of disease severity and containment stringency with individual life satisfaction

Dimension	Exposure or group	Est.	SE	Clust. p	Wild p
Age interaction	Stringency by baseline age	0.0011	0.0006	0.055	0.038
Age group	Disease, age 44 or younger	-0.0563	0.0148	0.001	0.007
Age group	Disease, age 45 to 59	-0.0260	0.0267	0.341	0.504
Age group	Disease, age 60 or older	-0.0409	0.0258	0.126	0.080
Hukou	Stringency, nonagricultural	-0.0565	0.0151	0.001	0.003
Hukou	Stringency, agricultural	-0.0024	0.0157	0.882	0.865
Residence	Stringency, urban	-0.0215	0.0129	0.110	0.103
Residence	Stringency, rural	-0.0259	0.0209	0.227	0.223
Education	Stringency, lower secondary or above	-0.0378	0.0128	0.007	0.013
Education	Stringency, below lower secondary	0.0254	0.0206	0.230	0.183

Hukou and current residence capture different dimensions of social position. When respondents are grouped by actual place of residence, the containment estimates are -0.0215 for urban residents and -0.0259 for rural residents. Their difference is only 0.0044

with $p = 0.813$. Disease coefficients are also similar in direction, at -0.0317 and -0.0270 . The reported pattern is more consistent with a distinction by hukou status than by current residence, but it does not establish that institutional status itself accounts for the difference. Nonagricultural hukou is correlated with formal employment, social insurance arrangements, and dependence on urban labour and service markets, which could be relevant to experiences of mobility restrictions. These factors remain plausible explanations rather than identified mediators.

4.5.3 Education

Education shows a similar contrast in the containment association. Among respondents with at least lower secondary education, the containment coefficient is -0.0378 , with $SE = 0.0128$, $p = 0.007$, and wild $p = 0.013$. Among respondents with less education, the estimate is 0.0254 , with $SE = 0.0206$, $p = 0.230$, and wild $p = 0.183$. The reported education-group comparison is statistically significant at conventional levels before multiple-testing adjustment. Disease estimates remain negative in both education groups, with a less pronounced contrast. One possible explanation is that more educated respondents had greater exposure to disruption in formal-sector employment or restricted mobility, while remote-working capacity could be relevant in the opposite direction. The regression estimates different conditional slopes across education groups but does not identify the mechanism behind them.

The heterogeneity evidence is more informative when read as a pattern than as a list of isolated significant coefficients. Age, hukou, residence, education, income, health, and region generate many related tests, which raises the probability of chance findings. Formal between-group comparisons are needed to assess heterogeneity; a significant coefficient in one group and a non-significant coefficient in another are not sufficient. For hukou, the reported clustered between-group test and the wild bootstrap result for the nonagricultural subgroup support further investigation, but they are different tests, and a bootstrap result for the between-group contrast is not reported here. Education and age also provide suggestive evidence, although these dimensions overlap with employment and hukou. As false discovery rate-adjusted results are not reported in this section, I treat the subgroup findings as exploratory evidence for further investigation rather than a definitive ranking of vulnerability.

Precision also differs sharply across subgroup models. Some categories contain many respondents but still rely on a small number of independent provincial exposure units, while regional models can contain as few as three clusters. A large number of individual observations cannot replace independent exposure variation. This distinction helps explain why conventional regional significance findings receive much weaker support from the reported wild bootstrap tests. Subgroups defined at the individual level may retain more provinces than regional subsamples, but their estimates also require caution. Group sample size, the number of represented provinces, and confidence intervals are therefore central to interpreting both statistical and substantive uncertainty.

4.5.4 Geographical Distribution of the Estimated Burden

Region specific interaction coefficients are too imprecise for formal causal comparison. Conventional clustered tests suggest negative disease coefficients in central and north eastern China, but wild bootstrap p values rise to 0.274 and 0.265 because the two regions contain only six and three sampled provinces. I therefore describe geographical variation by combining the common full sample coefficients with observed provincial doses. The calculation is a model implied decomposition, not a new region specific treatment effect estimate. It assumes common conditional slopes and attributes provincial differences to unequal exposure intensity.

The descriptive decomposition is defined as

$$\widehat{\Delta LS}_p = \hat{\beta}_D Disease_p + \hat{\beta}_L Stringency_p.$$

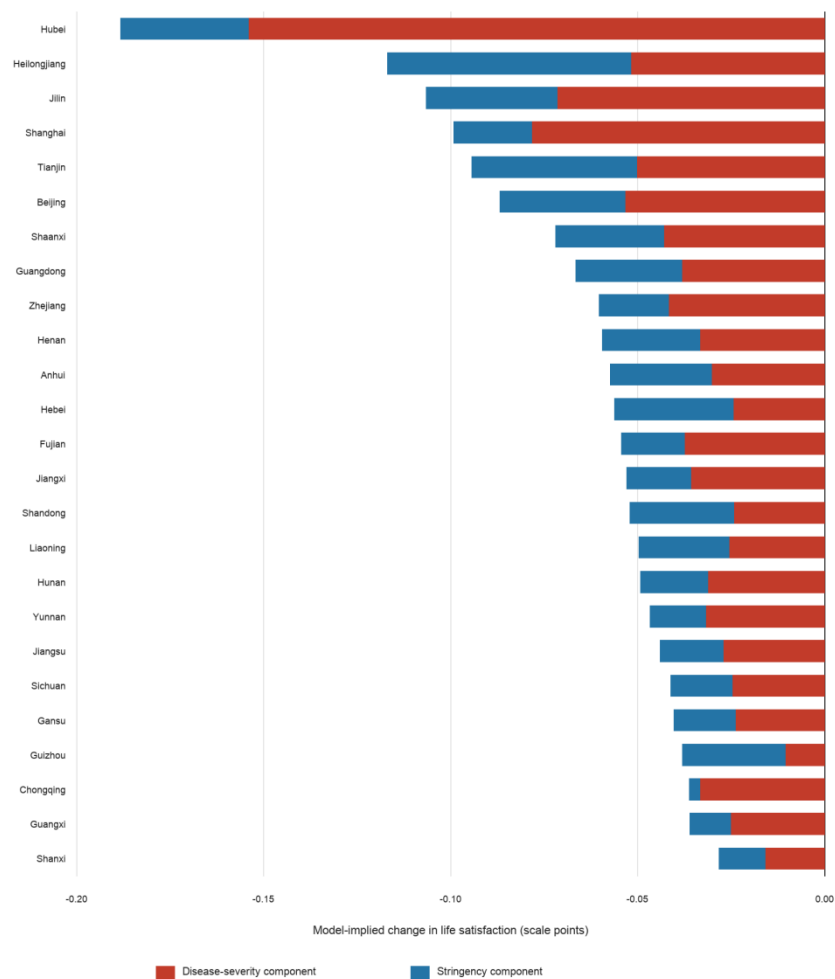


Figure 2 Model implied provincial burden from common disease and containment coefficients

The model-implied combined difference is most negative in Hubei at -0.188 points, followed by Heilongjiang at -0.117 , Jilin at -0.106 , Shanghai at -0.099 , Tianjin at -0.094 , and Beijing at -0.087 . The disease component forms the larger part of Hubei's model-implied value, whereas the strict containment component represents almost half of the value in Heilongjiang. Shanxi has the smallest combined magnitude at -0.028 , followed by Guangxi and Chongqing at about -0.036 and Guizhou at -0.038 . Weighted regional averages are -0.073 in the north east, -0.066 in the east, -0.062 in central China, and -0.043 in the west. These point calculations combine common estimated slopes with unequal observed exposures, and their interpretation inherits the uncertainty in those slopes, particularly the containment coefficient. They are not direct measures of provincial welfare loss and do not establish that residents in one region are more sensitive to a given exposure level.

4.6 Summary of Findings

The results address the four research questions with different levels of certainty.

Accumulated disease severity is negatively associated with life satisfaction in the preferred model, with comparatively precise inference, but the strength of this finding depends on the exposure definition and provincial covariate adjustment. Strict containment duration also has a negative point estimate, yet the available design cannot distinguish the average coefficient from zero with sufficient precision. The two standardised coefficients are not statistically different, so the data do not establish a ranking of their underlying association magnitudes. Wave-specific point estimates suggest a more negative disease association in 2020 and a more negative containment association in 2022, but neither formal between-wave comparison is significant at the 5% level. Exploratory heterogeneity patterns are most apparent in the disease estimates for younger respondents and the containment estimates for nonagricultural hukou holders and respondents with more education. These findings remain subject to multiple-testing concerns and do not support a general claim of greater vulnerability among urban residents. The geographical decomposition illustrates unequal exposure under common coefficients, without establishing regional differences in sensitivity or causal welfare losses. Overall, the section supports a cautious interpretation of uneven conditional associations with life satisfaction across China, with stronger evidence for accumulated disease severity than for the average duration of strict containment.

5 Conclusion

5.1 Summary of the study

This thesis examines whether two distinct dimensions of the COVID-19 shock—disease severity and containment stringency—were associated with changes in individual life satisfaction in China. Existing research often treats the pandemic as a single shock or relies on national post-outbreak indicators, making it difficult to distinguish epidemic conditions from policy responses or to benchmark changes against a pre-pandemic baseline. To address this limitation, the study links seven waves of the China Family Panel Studies (CFPS), covering 2010–2022, with province-level data from the Oxford COVID-19 Government Response Tracker.

The analytical sample comprises 17,733 person-wave observations from 3,825 respondents across 25 provinces. Disease exposure is measured as the logarithm of the mean cumulative confirmed-case rate within each wave-level exposure window, while containment exposure is measured as the number of months in which the provincial Stringency Index reached at least 70. The continuous-intensity difference-in-differences specification combines individual fixed effects with region-by-wave fixed effects. Because all provinces experienced the pandemic and exact interview dates are unavailable, the estimates are interpreted as conditional associations rather than independent causal effects. Province-clustered standard errors and a wild cluster bootstrap are used to characterise statistical uncertainty.

In the preferred specification, higher disease severity is associated with lower life satisfaction ($\beta = -0.0331$, $SE = 0.0117$, clustered $p = 0.009$; wild $p = 0.019$). The evidence is strongest when disease burden is measured using logged cumulative confirmed cases, whereas alternative disease definitions and covariate specifications yield less stable estimates. The result therefore supports a negative conditional association under the preferred specification, rather than a definitive causal effect.

Containment duration also has a negative point estimate ($\beta = -0.0172$, $p = 0.213$), but the confidence interval includes zero; the study therefore does not identify a statistically precise average association. Exploratory subgroup analyses indicate more negative containment coefficients among respondents with non-agricultural hukou and higher education, whereas the urban-rural contrast is close to zero. Age-group differences are also observed in the disease relationship, but they do not follow a simple monotonic

gradient. Given multiple testing and limited independent exposure variation, these subgroup patterns should be treated as exploratory.

The data likewise do not support a statistical ranking of the two channels: the standardised difference is -0.0091 ($p = 0.627$). Wave-specific point estimates suggest a more negative disease association in 2020 and a more negative containment association in 2022, but formal between-wave comparisons remain imprecise. The model-implied provincial decomposition places Hubei at the largest combined difference (-0.188); this reflects observed exposure levels under common coefficients, rather than greater provincial sensitivity or a direct estimate of causal welfare loss.

5.2 Theoretical and practical contributions

The principal theoretical contribution is to distinguish epidemic conditions from containment policy within a common empirical framework. Similar point estimates do not imply equally strong evidence: the preferred specification provides stronger support for a negative disease association than for an average containment association. Treating COVID-19 as a single undifferentiated event would obscure this distinction, although the joint model does not identify two independent total causal effects.

The panel structure also changes the basis of comparison. Repeated observations of the same respondents allow pandemic-period changes to be evaluated against an individual pre-pandemic history, rather than relying solely on cross-sectional differences between people observed during the crisis. This strengthens the temporal basis of the analysis, although any stronger causal interpretation remains contingent on the identifying assumptions and data limitations set out in the methodology.

The heterogeneity analysis further illustrates the importance of separating supported findings, exploratory subgroup patterns and possible mechanisms. Age and region are the focal heterogeneity dimensions, whereas hukou, residence and education comparisons are exploratory. In particular, the near-zero urban-rural difference does not support a distinct conclusion for urban residents. Potential explanations involving labour-market exposure, mobility constraints, adaptation or cumulative fatigue therefore remain hypotheses rather than mechanisms identified by the models.

The practical implications should consequently be interpreted with caution. The negative association between disease burden and life satisfaction indicates that subjective

wellbeing is relevant when assessing pandemic conditions. However, the imprecise average containment estimate and exploratory subgroup evidence do not justify targeting support specifically at urban residents. More defensible implications are to monitor wellbeing during prolonged restrictions, maintain broadly accessible income, employment and mental-health support where disruption is greatest, and communicate policy duration and exit conditions clearly.

5.3 Limitations

The main limitation arises from province-level exposure assignment, which leaves only 25 independent provincial units for inference. Province-level measures also conceal substantial local and individual variation; confirmed case counts depend partly on testing and reporting practices; and the Stringency Index threshold is researcher-defined. In addition, exact interview dates are unavailable, meaning that measured exposure may postdate the outcome for some respondents.

Disease burden and containment policy are also jointly determined: policy responds to case levels and may in turn affect subsequent cases. Mutual adjustment may therefore block part of the policy pathway or introduce simultaneity bias, so the joint model cannot identify two independent total causal effects. The disease result is also sensitive to exposure definition and covariate adjustment, while subgroup analyses are subject to multiple-testing concerns and some regional estimates rely on very few clusters.

Finally, the biennial and unbalanced CFPS panel cannot precisely distinguish immediate reactions from adaptation, recovery or cumulative fatigue. Evidence from mainland China also cannot be assumed to generalise automatically to other institutional or policy settings. Taken together, these limitations mean that the findings are best interpreted as conditional panel associations rather than definitive causal estimates.

5.4 Future research directions

Future research could bring exposure measurement closer to the individual by linking infection records, quarantine histories, mobility data or high-frequency local policy records to wellbeing outcomes. Research designs based on more precisely timed local policy changes could also improve temporal ordering and increase treatment variation.

Direct survey measures of infection, quarantine, income interruption, mobility restrictions and social contact would help distinguish province-level exposure from restrictions actually experienced by individuals and would permit more direct tests of proposed mechanisms. Future studies should also pre-specify alternative exposure definitions and subgroup comparisons and report the full range of sensitivity analyses.

Higher-frequency panel data could distinguish immediate wellbeing changes from subsequent adaptation and test cumulative-fatigue explanations more directly. Replication across countries with different policy settings, combined with measures of employment disruption, perceived health risk, social contact and trust, would help assess external validity and clarify the pathways associated with pandemic exposure.

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Appendices

Appendix A: List of 25 Core Provinces

Table 4 List of 25 Core Provinces

No.	Province	OxCGRT Region Code
1	Shanghai	CHN_SH
2	Yunnan	CHN_YN
3	Beijing	CHN_BJ
4	Jilin	CHN_JL
5	Sichuan	CHN_SC
6	Tianjin	CHN_TJ
7	Anhui	CHN_AH
8	Shandong	CHN_SD
9	Shanxi	CHN_SX
10	Guangdong	CHN_GD
11	Guangxi	CHN_GX
12	Jiangsu	CHN_JS
13	Jiangxi	CHN_JX
14	Hebei	CHN_HE
15	Henan	CHN_HA
16	Zhejiang	CHN_ZJ
17	Hubei	CHN_HB
18	Hunan	CHN_HN
19	Gansu	CHN_GS
20	Fujian	CHN_FJ
21	Guizhou	CHN_GZ
22	Liaoning	CHN_LN
23	Chongqing	CHN_CQ
24	Shaanxi	CHN_SN
25	Heilongjiang	CHN_HL

Appendix B: Ethics Application Form: Student Research

Anyone conducting research under the auspices of the Institute (staff, students or visitors) where the research involves human participants or the use of data collected from human participants, is required to gain ethical approval before starting. This includes preliminary and pilot studies. Please answer all

relevant questions in terms that can be understood by a lay person and note that your form may be returned if incomplete.

For further support and guidance please see accompanying guidelines and the Ethics Review Procedures for Student Research or contact your supervisor.

Before completing this form you will need to discuss your proposal fully with your supervisor(s).

Please attach all supporting documents and letters.

For all Psychology students, this form should be completed with reference to the British Psychological Society (BPS) Code of Human Research Ethics and Code of Ethics and Conduct.

Section 1: Project Details

- a) Project title: [An Empirical Study of Factors Influencing Subjective Well-being Across Diverse National Populations in the Post-Pandemic Era](#)
- b) Student name: [REDACTED]
- c) Supervisor/Personal Tutor: [REDACTED]
- d) Department: [Social Research Institute](#)
- e) Course category (tick one):
 - PhD/MPhil ☐
 - EdD ☐
 - MRes ☐
 - DEdPsy ☐
 - MTeach ☐
 - MA/MSc ☒
 - ITE ☐
 - Diploma (state which) ☐ Enter text
 - Other (state which) ☐ Enter text
- f) Course/module title: [Social Research Methods with Data Science MSc](#)
- g) If applicable, state who the funder is and if funding has been confirmed: [N/A](#)
- h) Intended research start date: [01/01/2026](#)
- i) Intended research end date: [01/08/2026](#)
- j) Country fieldwork will be conducted in: [The United Kingdom](#)

If research to be conducted abroad please check the [Foreign and Commonwealth Office \(FCO\)](#) and submit a completed travel risk assessment form (see guidelines). If the FCO advice is against travel this will be required before ethical approval can be granted: [UCL travel advice webpage](#)

- k) Has this project been considered by another (external) Research Ethics Committee?

Yes ☐

External Committee Name:

Date of Approval:

No ☒ **go to Section 2**

If yes:

- Submit a copy of the approval letter with this application.
- Proceed to Section 10 Attachments.

Note: Ensure that you check the guidelines carefully as research with some participants will require ethical approval from a different ethics committee such as the [National Research Ethics Service](#) (NRES) or [Social Care Research Ethics Committee](#) (SCREC). In addition, if your research is based in another institution then you may be required to apply to their research ethics committee.

Section 2 - Research methods summary (tick all that apply)

- ☐ Interviews
- ☐ Focus Groups
- ☐ Questionnaires
- ☐ Action Research
- ☐ Observation
- ☐ Literature Review
- ☐ Controlled trial/other intervention study
- ☐ Use of personal records
- ☐ Systematic review – **if only method used go to Section 5**
- ☒ Secondary data analysis – **if secondary analysis used go to Section 6**
- ☐ Advisory/consultation/collaborative groups
- ☐ Other, give details:

Please provide an overview of the project, focusing on your methodology. This should include some or all of the following: purpose of the research, aims, main research questions, research design, participants, sampling, data collection (including justifications for methods chosen and description of topics/questions to be asked), reporting and dissemination. Please focus on your methodology; the theory, policy, or literary background of your work can be provided in an attached document (i.e. a full research proposal or case for support document). *Minimum 150 words required.*

The study aims to investigate the principal factors influencing subjective well-being (SWB) among global residents in the post-pandemic era. It will investigate whether economic factors, such as GDP, are the most significant variables affecting residents' subjective well-being. This study draws on the World Happiness Report (WHR) and uses secondary data for analysis. A range of indicators related to happiness will be included in the analytical framework, and multiple regression and panel data models are expected to be applied to identify several key factors affecting subjective well-being. In addition, a comparative analysis will be conducted across different countries. The research questions include, but are not limited to, the following: which social factors have the strongest explanatory power for subjective well-being, and what differences can be observed in the subjective well-being of residents across countries. The findings will eventually be summarized and presented in a dissertation of approximately 12,000 words.

Section 3 – research Participants (tick all that apply)

a) Will your research involve human participants?

Yes ☐

No ☒ (if 'No', go to Section 4)

b) Who are the participants for this project (i.e. what sorts of people will be involved)? Tick all that apply

☐ Early years/pre-school

☐ Ages 5-11

☐ Ages 12-16

☐ Young people aged 17-18

☐ Adults please specify below

☐ Unknown – specify below

☐ No participants

Enter text

Note: Ensure that you check the guidelines carefully as research with some participants will require ethical approval from a different ethics committee such as the [National Research Ethics Service](#) (NRES) or [Social Care Research Ethics Committee](#) (SCREC).

- c) If participants are under the responsibility of others (such as parents, teachers or medical staff) how do you intend to obtain permission to approach the participants to take part in the study? *(Please attach approach letters or details of permission procedures - see Section 9)*

Enter text

- d) How will participants be recruited (identified and approached)?

Enter text

- e) Describe the process you will use to inform participants about what you are doing

Enter text

- f) How will you obtain the consent of participants? Will this be written? How will it be made clear to participants that they may withdraw consent to participate at any time?

Enter text

- g) **Studies involving questionnaires:** will participants be given the option of omitting questions they do not wish to answer?

Yes ☐

No* ☐

***If no**, please explain why below, and ensure that you cover any ethical issues arising from this in Section 8

Enter text

- h) **Studies involving observation:** please confirm whether participants will be asked for their informed consent to be observed

Yes ☐

No* ☐

***If no**, read the guidelines (Ethical Issues section) and explain why below and ensure that you cover any ethical issues arising from this in section 8

Enter text

- i) Might participants experience anxiety, discomfort, or embarrassment as a result of your study?

Yes* ☐

No* ☐

***If yes**, what steps will you take to explain and minimise this?

Enter text

***If no**, explain how you can be sure that no discomfort or embarrassment will arise?

Enter text

- j) Will your project involve deliberately misleading participants (deception) in any way?

Yes* ☐

No ☐

If yes, please provide further details below and ensure that you cover any ethical issues arising from this in Section 8

Enter text

- k) Will you debrief participants at the end of their participation (i.e. give them a brief explanation of the study)?

Yes ☐

No* ☐

***If no**, please explain why below and ensure that you cover any ethical issues arising from this in section 8

Enter text

- l) Will participants be given information about the findings of your study? (This could be a brief summary of your findings in general; it is not the same as an individual debriefing)

Yes ☐

No* ☐

If no, why not?

Enter text

Section 4 - Security-sensitive material (only complete if applicable)

Security sensitive research includes: commissioned by the military; commissioned under an EU security call; involves the acquisition of security clearances; concerns terrorist or extreme groups.

- a. Will your project consider or encounter security-sensitive material?
Yes* ☐ No ☒
- b. Will you be visiting websites associated with extreme or terrorist organisations?
Yes* ☐ No ☒
- c. Will you be storing or transmitting any materials that could be interpreted as promoting or endorsing terrorist acts?
Yes* ☐ No ☒

* Give further details in **Section 8 Ethical Issues**

Section 5 – Systematic reviews of research (only complete if applicable)

- a. Will you be collecting any new data from participants?
Yes* ☐ No ☐
- b. Will you be analysing any secondary data?
Yes* ☐ No ☐

* Give further details in **Section 8 Ethical Issues**

*If your methods do not involve engagement with participants (e.g. systematic review, literature review) **and** if you have answered **No** to both questions, please go to **Section 10 Attachments**.*

Section 6 - Secondary data analysis (only complete if applicable)

- a. Name of dataset/s: [World Happiness Report](#)
- b. Owner of dataset/s: [the Gallup World Poll](#)
- c. Are the data in the public domain?
Yes ☒ No ☐

If no, do you have the owner's permission/license?

Yes ☐ No* ☐

- d. Are the data special category personal data (i.e. personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation)?

Yes* ☐ No ☒

e. Will you be conducting analysis within the remit it was originally collected for?

Yes ☒ No* ☐

f. **If no**, was consent gained from participants for subsequent/future analysis?

Yes ☐ No* ☐

g. **If no**, was data collected prior to ethics approval process?

Yes ☐ No* ☐

* Give further details in **Section 8 Ethical Issues** ☐

*If secondary analysis is only method used **and** no answers with asterisks are ticked, go to **Section 9 Attachments**.*

Section 7 – Data storage and security

a) Confirm that all personal data will be stored and processed in compliance with the General Data Protection Registration (GDPR). (See the [Guidelines](#) and the Institute's Data Protection & Records Management Policy for more detail.)

Yes ☒

b) Will personal data be processed or be sent outside of the European Economic Area (EEA)?

Yes* ☐

No ☒

***If yes**, please confirm that there are adequate levels of protections in compliance with the DPA 2018 and state what these arrangements are below

Enter text

c) Who will have access to the data and personal information, including advisory/consultation groups, and during transcription?

[Only the researcher](#)

During the research

d) Where will the data be stored?

[OneDrive](#)

e) Will mobile devices such as USB storage and laptops be used?

Yes ☒

No ☐

***If yes**, state what mobile devices will be used

[Laptop](#)

After the research

f) Where will the data be stored?

OneDrive

g) How long will the data and records be kept for, and in what format?

Enter text

h) Will the data be archived for use by other researchers?

Yes ☐

No ☒

***If yes**, please provide details

Enter text

Section 8 – Ethical Issues

Are there particular features of the proposed work which may raise ethical concerns or add to the complexity of ethical decision making? If so, please outline how you will deal with these below.

It is important that you demonstrate your awareness of potential risks or harm that may arise as a result of your research. You should then demonstrate that you have considered ways to minimise the likelihood and impact of each potential harm that you have identified. Please be as specific as possible in describing the ethical issues you will have to address. Please consider / address ALL issues that may apply from the below.

Ethical concerns may include, but not be limited to, the following areas:

- Methods
- Sampling
- Recruitment
- Gatekeepers
- Informed consent
- Potentially vulnerable participants
- Safeguarding/child protection
- Sensitive topics
- International research
- Risks to participants and/or researchers
- Confidentiality/Anonymity
- Disclosures/limits to confidentiality
- Data storage and security both during and after the research (including transfer, sharing, encryption, protection)
- Reporting
- Dissemination and use of findings

This study uses a publicly available secondary dataset, and no primary data collection, participant recruitment, or direct interaction with individuals is involved. No individual identification is possible from the data, and the dataset does not contain special category personal data. As a result, the ethical risk associated with this project is minimal. Data will be stored securely in line with UCL policy, including the use of password-protected and access-controlled storage where appropriate. Findings will be reported only in aggregate form, and no attempt will be made to identify any individual or disclose confidential information.

Section 9 – Further information

Outline any other information you feel is relevant to this submission, using a separate sheet or attachments if necessary

N/A

Section 10 – Face to Face data collection (only complete if you will be collecting face to face data outside of the UK)

Provide details of your back-up plan for online data collection methods or secondary data analysis, which will be put into action should you need to cease F2F data collection (max 150 words).

Please note that should you need to change your data collection methods due to a change in Covid19 regulations in the country context of your research (for example, if lockdown is re-introduced) you will need to submit to your supervisor:

- a) a revised information sheet
- b) consent form
- c) details of any additional/different risks linked to online data collection.

In cases where back-up plans have to be implemented, students must not proceed with online data collection until they have received written approval (via email) from their tutor/supervisor.

Enter text

Section 11 – Attachments

Please attach the following items to this form, or explain if not attached:

- a) Information sheets and other materials to be used to inform potential participants about the research, including approach letters

Yes ☐ No ☐ N/A ☒

- b) Consent form

Yes ☐ No ☐ N/A ☒

- c) The proposal for the project

Yes ☒ No ☐ N/A ☐

d) Approval Letter from external Research Ethics Committee

Yes ☐ No ☐ N/A ☒

e) Full risk assessment

Yes ☐ No ☐ N/A ☒

Section 12 – Declaration

I have read, understood, and will abide by the following set of guidelines:

Yes ☒ No ☐

Please select guidelines this project will be abiding by:

BPS ☐

BERA ☐

BSA ☒

Other (please state) ☐ Enter text

I have discussed the ethics issues relation to my research with my supervisor

Yes ☒ No ☐

I have attended the appropriate ethics training provided by my course

Yes ☒ No ☐

I confirm that to the best of my knowledge:

The above information is correct and this is a full description of the ethics issues that may arise in the course of this project

Name: [Anonymous](#)

Date: [10/03/2026](#)

Once complete, please submit your complete ethics forms to your supervisor

Notes & References

Useful links & information

Professional code of ethics

You should read and understand relevant ethics guidelines, for example:

[British Psychological Society](#) (2021) *Code of Ethics and Conduct*

or

[British Educational Research Association](#) (2024) *Ethical Guidelines*

or

[British Sociological Association](#) (2017) *Statement of Ethical Practice*

Please see the respective websites for these or later versions; direct links to the latest versions are available on the [Institute of Education Research Ethics webpage](#)

Disclosure and Barring Service checks

If you are planning to carry out research in regulated Education environments such as Schools, or if your research will bring you into contact with children and young people (under the age of 18), you will need to have a Disclosure and Barring Service (DBS) CHECK, before you start. The DBS was previously known as the Criminal Records Bureau (CRB)). If you do not already hold a current DBS check, and have not registered with the DBS update service, you will need to obtain one through UCL.

Ensure that you apply for the DBS check in plenty of time as will take around 4 weeks, though can take longer depending on the circumstances.

Further references

Robson, Colin (2011). *Real world research: a resource for social scientists and practitioner researchers* (3rd edition). Oxford: Blackwell.

This text has a helpful section on ethical considerations.

Alderson, P. and Morrow, V. (2011) *The Ethics of Research with Children and Young People: A Practical Handbook*. London: Sage.

This text has useful suggestions if you are conducting research with children and young people.

Wiles, R. (2013) *What are Qualitative Research Ethics?* Bloomsbury.

A useful and short text covering areas including informed consent, approaches to research ethics including examples of ethical dilemmas.

Departmental use

If a project raises particularly challenging ethics issues, or a more detailed review would be appropriate, the supervisor must refer the application to the Research Development Administrator via email so that it can be submitted to the IOE Research Ethics Committee for consideration. A departmental research ethics coordinator or representative can advise you, either to support your review process, or help decide whether an application should be referred to the REC. If unsure please refer to the guidelines explaining when to refer the ethics application to the IOE Research Ethics Committee, posted on the committee's website.

Student name: [REDACTED]

Course: [MSc Social Research Methods with Data Science](#)

Project Title: [An Empirical Study of Factors Influencing Subjective Well-being Across Diverse National Populations in the Post-Pandemic Era](#)

Reviewer 1

Supervisor name: [REDACTED]

Supervisor comments:

[In my view, this research has negligible risks.](#)

Supervisor/first reviewer signature: [REDACTED]

Date: [11/03/2026](#)

Reviewer 2

Advisory committee/course team member name: [REDACTED]

Advisory committee/course team member comments:

[I see no ethical concerns with the planned research](#)

Advisory committee/course team member signature: [REDACTED]

Date: [23/04/2026](#)

Decision on behalf of reviewers

Approved ☒

Referred back to applicant and supervisor ☐

Referred to the REC for review ☐

Recording

Recorded in the student information system ☐

Once completed and approved, please send this form and associated documents to the relevant programme administrator to record on the student information system and to securely store.

Further guidance on ethical issues can be found on the UCL Institute of Education Research Ethics Committee website

Appendix C: Robustness and Sensitivity

C.1 Alternative Exposure Measures

Table 5 examines sensitivity to the measurement of disease burden and to covariate adjustment. In Panel A, both cumulative disease measures have negative coefficients with conventional province-clustered p values below 0.05: -0.0331 for the logged cumulative rate ($p = 0.009$) and -0.0012 for the raw cumulative rate ($p = 0.003$). The logged and raw new-case rates also have negative coefficients, -0.0144 ($p = 0.185$) and -0.0003 ($p = 0.231$), but provide less precise evidence of an association. Because these measures use different units and transformations, their coefficient magnitudes cannot be compared directly.

The negative association under both cumulative definitions suggests that this finding does not depend solely on taking logarithms. However, the weaker evidence for new-case rates shows that the result is sensitive to whether exposure represents accumulated burden or recent incidence. Cumulative measures retain information about earlier outbreaks and may therefore reflect a different exposure history. These comparisons do not establish that one measure identifies a causal effect or that the disease result is robust to all plausible definitions.

Containment exposure is measured as the number of months with a Stringency Index of at least 70. This is a researcher-selected threshold rather than an official policy boundary. Table 5 varies disease measurement and covariate adjustment; it does not test alternative definitions of containment exposure. Results for thresholds of 60 and 80, mean stringency, or individual policy components are not presented here. Robustness to those definitions therefore remains unestablished, and the containment findings should be interpreted specifically in relation to months above the selected threshold.

Table 5 Sensitivity of disease-severity and containment-stringency estimates to alternative exposure measures and covariate specifications

Panel	Specification	Disease est.	Disease p	Policy est.	Policy p
A, measure	Logged cumulative rate, preferred	-0.0331	0.009	-0.0172	0.213
A, measure	Raw cumulative rate	-0.0012	0.003	-0.0220	0.106
A, measure	Logged new case rate	-0.0144	0.185	-0.0204	0.167
A, measure	Raw new case rate	-0.0003	0.231	-0.0225	0.118
B, controls	No additional controls	-0.0132	0.336	-0.0205	0.179
B, controls	Provincial adjustment only	-0.0314	0.007	-0.0175	0.206
B, controls	Individual covariates only	-0.0143	0.325	-0.0206	0.182
B, controls	Full preferred model	-0.0331	0.009	-0.0172	0.213
B, controls	Observed post 2020 macro values	-0.0103	0.313	-0.0219	0.159

Note. All specifications use 17,733 person-wave observations. Panel A varies the disease exposure measure, whose units differ across rows. Panel B varies covariate adjustment while retaining the preferred exposure definitions. Est. denotes estimate. All p values in this table use conventional province-clustered inference, not wild cluster bootstrap inference.

C.2 Covariate and Specification Sensitivity

Panel B of Table 5 shows that disease estimates are sensitive to covariate adjustment. The coefficient is -0.0132 without additional controls ($p = 0.336$), -0.0314 with provincial adjustment only ($p = 0.007$), -0.0143 with individual covariates only ($p = 0.325$), and -0.0331 in the full model ($p = 0.009$). The negative sign persists, but the strength of the statistical evidence depends on the adjustment set. Provincial adjustment is therefore consequential for the main result and should not be treated as a minor modelling choice.

Replacing frozen 2018 macroeconomic values with observed post-2020 values yields a disease coefficient of -0.0103 ($p = 0.313$). This change could reflect adjustment for variables affected by the pandemic, but it does not establish that the baseline specification recovers the correct counterfactual. Across Panel B, containment coefficients range from -0.0219 to -0.0172 and none has $p < 0.05$. Their relatively stable magnitude does not establish the absence of an association, since the estimates remain statistically imprecise.

Region-by-wave fixed effects allow regional time paths to differ, but favourable pre-trend diagnostics do not by themselves establish identification or rule out time-varying confounding. The negative disease association in Table 5 is supported by both cumulative

exposure definitions. Overall, the findings remain sensitive to exposure definition and covariate adjustment.

C.3 Influential Provinces and Statistical Inference

The reported leave-one-province-out estimates remain negative for both exposures. Disease coefficients range from -0.0426 when Shanghai is excluded to -0.0264 when Jiangxi is excluded. Twenty-three of the 25 re-estimations have $p < 0.05$; the exceptions exclude Hubei ($p = 0.090$) and Jiangxi ($p = 0.052$). Containment coefficients range from -0.0337 to -0.0112 , with $p < 0.05$ in only one re-estimation. These results suggest that no single province is necessary for the negative signs, while the strength of the statistical evidence varies with the sample. Changes in p values alone cannot distinguish the contribution of an influential province from the loss of precision caused by removing a cluster.

Excluding the 2018 wave yields a disease coefficient of -0.0412 ($p = 0.006$) and a containment coefficient of -0.0123 ($p = 0.383$). This preserves the reported qualitative pattern, but it remains a check on one aspect of sample composition rather than a general validation of the model.

With only 25 province-level clusters, conventional inference warrants caution. Table 5 reports conventional province-clustered p values; it does not establish robustness under wild cluster bootstrap inference. The within-region permutation exercise yields $p = 0.097$ for disease severity. Since exposure was not randomly assigned, this is a diagnostic rather than an exact randomisation test. City-level clustering cannot replace assessment of dependence at the province level, where exposures are assigned. Overall, disease estimates are more consistent in sign than in statistical significance, while containment results remain inconclusive.